Slow Light in Optical Waveguides

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As evidenced by this Volume, there has been a flurry of activity over the last decade on tailoring the dispersive properties of optical materials [1]. What has captured the attention of the research community were some of the early results on creating spectral regions of large normal dispersion [2–4]. Large normal dispersion results in extremely small group velocities, where the group velocity is the approximate speed at which a pulse of light propagates through a material. We denote the group velocity by $\upsilon_g = c/n_g$, where c is the speed of light in vacuum and n_g is known as the group index. In the early experiments, described in greater detail in Chapter *, a dilute gas of atoms is illuminated by a "control" or "coupling" beam whose frequency is tuned precisely to an optical transition of an atom. This control field modifies the absorption and dispersion properties of another atomic transition that share a common energy level. A narrow transparency window is created on this second transition – a process known as electromagnetically induced transparency (EIT) – and, within this window, υ_g takes on extremely small values. Many experiments have now observed $\upsilon_g \sim 1$ m/s or less, implying $n_g > 10^8$. This result is remarkable considering that fact that the refractive index n of a material rarely exceeds 3 in the visible part of the spectrum. What are the implications of such large group indices? What applications are enabled by this basic science discovery?

One immediate application that comes to mind is to use slow light (the situation where $\upsilon_g \ll c$) for realizing a real-time adjustable buffer for optical pulses. A buffer that is capable of delaying an entire packet of optical information can substantially increase the efficiency of routers in optical telecommunication networks [5–8], for example. The primary motivation for our research over the past few years has been to develop new mechanisms for realizing slow light that operate at or near room temperature and occur in an optical waveguide. The waveguide geometry allows light-matter interactions to take place over long distances where the transverse dimension of the light is of the order of the wavelength, thereby lowering the optical power needed to create the slow-light effect. Also, a waveguide-based slow-light device can be compact and integrated with existing technologies.

The primary goal of this Chapter is to review our own research on slow light in optical waveguides. In particular, we describe how slow light can be achieved by stimulated Brillouin and stimulated Raman scattering (SBS and SRS, respectively) in transparent optical fibers, by coherent population oscillations (CPO) in erbium-doped fiber amplifiers, by EIT in gas-filled hollow-core fibers, and by wavelength conversion and fiber dispersion.

I. SLOW LIGHT VIA STIMULATED SCATTERING

To understand how slow light can be achieved via stimulated scattering, it is important to recall the Kramers-Kronig relations [9], which relate the real and imaginary part of the complex refractive index of a causal dielectric. In particular, frequency-dependent material absorption (or gain) is necessarily associated with a frequency-dependent refractive index. Considering the definition for the group index, which is given by $n_g = n + \omega dn/d\omega$, where ω is the optical frequency, we see that the group index differs from the refractive index by the so-called dispersive term $\omega dn/d\omega$. Thus, large values of n_g are obtained when there is a substantial change in refractive index over a narrow frequency interval (making $dn/d\omega$ large), which is associated with a rapid spectral variation in the absorption of the material.

In simulated scattering processes [10], light scattering occurs as a result of highly localized changes in the dielectric constant of a medium. For sufficiently strong light fields, these changes can be induced via coupling of a material excitation to two light fields whose difference in frequencies is given by the frequency of the excitation. The excitation gives rise to a nonlinear optical coupling between the fields, which allows power to flow from one beam to another and which can give rise to absorption or amplification of a probe beam. This coupling occurs over a narrow spectral range, which gives rise to a narrow resonance that can be used to control v_g by adjusting the laser beam intensities. Important features of a stimulated scattering resonance is that it can occur at room temperature and it is induced by a pump laser beam and hence can occur over the entire range of frequencies where the material is transparent. In our research, we have investigated the material resonances arising from both stimulated Brillouin scattering (SBS) and stimulated Raman scattering (SRS), as described below.

A. Slow light via SBS

In the SBS process, a high-frequency acoustic wave is induced in the material via electrostriction for which the density of a material increases in regions of high optical intensity. The process of SBS can be described classically as a nonlinear interaction between the pump (at angular frequency ω_p) and a probe field (ω) through the induced acoustic wave (Ω_B) [10]. The acoustic wave in turn modulates the refractive index of the medium and scatters pump light into (out of) the probe wave when its frequency is downshifted (upshifted) by the acoustic frequency. This process leads to a strong coupling between the three waves when this resonance condition is satisfied, which results in exponential amplification (absorption) of the probe wave. Efficient SBS occurs when both energy and momentum are conserved, which is satisfied when the pump and probe waves counterpropagate.

Slow light due to SBS can be understood by the studying the resonances experienced by the probe wave for the case when the medium is pumped by a continuous-wave beam. Here, we focus on the case when the frequency of a counterpropagating probe beam is near the amplifying resonance (also known as the Stokes resonance); the results are generalized straightforwardly to the case of the absorbing (or anti-Stokes resonance). In the small-signal limit (*i.e.*, pump depletion is negligible), the probe wave in the fiber (propagating in $+z$ direction) experiences an effective complex refractive index $\tilde{n}(\omega)$ given by

$$\tilde{n} = n_f - i\frac{c}{\omega}\frac{g_0 I_p}{1 - i2\delta\omega/\Gamma_B}, \tag{1}$$

where n_f is the modal index of the fiber mode, I_p is the pump intensity, g_0 is line-center gain factor, $\delta\omega = \omega - \omega_p + \Omega_B$, and $\Gamma_B/2\pi$ is the FWHM (full width at half maximum) linewidth of the Brillouin resonance. For a typical optical telecommunication fiber, $\Omega_B/2\pi \sim 10$ GHz and $\Gamma_B/2\pi \sim 30$ MHz. The fact that the resonance linewidth is so narrow – comparable to the natural linewidth of atomic transitions used in atom-based slow-light – suggests controlling v_g in an optical fiber is achievable. From Eq. (1), it is seen that the probe wave experiences gain and dispersion in the form of a Lorentzian-shaped resonance. The gain coefficient

$g = -2(\omega/c)\mathrm{Im}(\tilde{n})$, real refractive index $n = \mathrm{Re}(\tilde{n})$, and group index $n_g = n + \omega(dn/d\omega)$ are given by

$$g(\omega) = \frac{g_0 I_p}{1 + 4\delta\omega^2/\Gamma_B^2}, \tag{2a}$$

$$n(\omega) = n_f + \frac{c g_0 I_p}{\omega} \frac{\delta\omega/\Gamma_B}{1 + 4\delta\omega^2/\Gamma_B^2}, \tag{2b}$$

$$n_g(\omega) = n_{fg} + \frac{c g_0 I_p}{\Gamma_B} \frac{1 - 4\delta\omega^2/\Gamma_B^2}{(1 + 4\delta\omega^2/\Gamma_B^2)^2}, \tag{2c}$$

respectively, where n_{fg} is the group index of the fiber mode when SBS is absent. Figure 1 shows the refractive index, gain and group index for the SBS amplifying resonance. It is seen that normal dispersion near the center of the resonance leads to an increase in the group index and therefore a decrease in group velocity $\upsilon_g = c/n_g$.

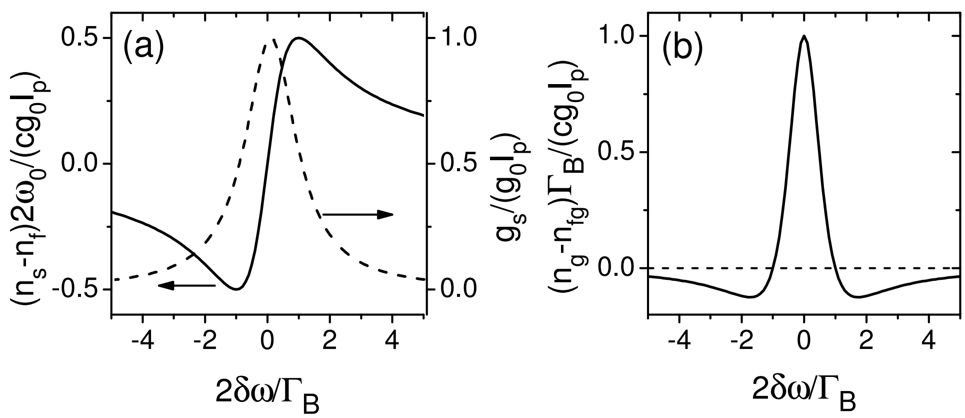

FIG. 1: Large dispersion of the SBS resonance. (a) Gain (solid line) and refractive index (dashed line) of the resonance. (b) Normalized group index of the resonance. (After Ref. [11].)

For the optical data buffering application mentioned above, one important characteristic of a slow-light device is its ability to controllably delay a pulse. For the situation where the majority of the pulse spectrum falls within the region where υ_g is nearly constant, the slow-light delay (defined as the difference between the transit times with and without SBS) is given by

$$\begin{aligned} \Delta T_d =& \frac{G}{\Gamma_B} \frac{1 - 4\delta\omega^2/\Gamma_B^2}{(1 + 4\delta\omega^2/\Gamma_B^2)^2} \\ \simeq& \frac{G}{\Gamma_B}(1 - 12\delta\omega^2/\Gamma_B^2) \quad \text{when} \quad 4\delta\omega^2/\Gamma_\mathrm{B}^2 \ll 1, \end{aligned} \tag{3}$$

where $G = g_0 I_p L$ is the gain parameter whose exponential e^G is the small-signal gain, and L is the fiber length. The maximum delay occurs at the peak of the Brillouin gain ($\delta\omega = 0$) and is given simply by

$$\Delta T_d = G/\Gamma_B. \tag{4}$$

It is seen that the slow light delay ΔT_d is tunable by adjusting the pump intensity. Equation (4) gives $\Delta T_d \simeq 1.2$ ns/dB for $\Gamma_B/2\pi = 30$ MHz.

Slow light delay is always accompanied by some degree of pulse distortion due to the fact that some portion of the pulse spectrum extends to regions where there is substantial variation in the gain and group index. For example, a Gaussian-shaped pulse will be temporally broadened by a factor [12]

$$B \equiv \tau_{out}/\tau_{in} = \left[1 + (16 \ln 2)G/\tau_{in}^2 \Gamma_B^2\right]^{1/2}, \tag{5}$$

where τ_{in} and τ_{out} are the pulse widths (FWHM) of the input and output pulses, respectively, and where third and higher-order dispersion are neglected in obtaining this expression. Pulse-width broadening is mainly due to the spectral reshaping arising from the bandwidth-limited gain, which narrows the spectral width of the output pulse.

Fractional delay (the ratio of delay to the pulse width) is related to the pulse broadening by

$$\Delta T_d/\tau_{in} = \left[(B^2 - 1)G/(16 \ln 2)\right]^{1/2}. \tag{6}$$

This result suggests that there exists a maximum achievable fractional delay in a single SBS slow-light element because G is limited due to self generation. In the self-generation process, spontaneously scattered light produced at the Stokes frequency (generated by thermal fluctuations) and near the entrance face of the fiber experiences exponential growth due to the SBS process. For sufficiently high gain (typically $G \sim 10 - 15$ for long lengths of optical fiber), the self-generated light depletes the pump beam leading to gain saturation [11]. Using a limit of $G = 15$ and a constraint of $B = 2$ gives a fractional delay limit of 2.6 for a single Lorentzian amplifying resonance. The issue of self-generation can be avoided using multiple SBS slow-light delay lines separated by attenuators as demonstrated by Song *et al.* [13].

Tunable slow-light delay via SBS in an optical fiber was first demonstrated independently by Song *et al.* [14] and Okawachi *et al.* [12]. In the experiment by Okawachi *et al.* (see Fig. 2), a single continuous-wave narrow-linewidth tunable laser at 1550 nm was used to generate

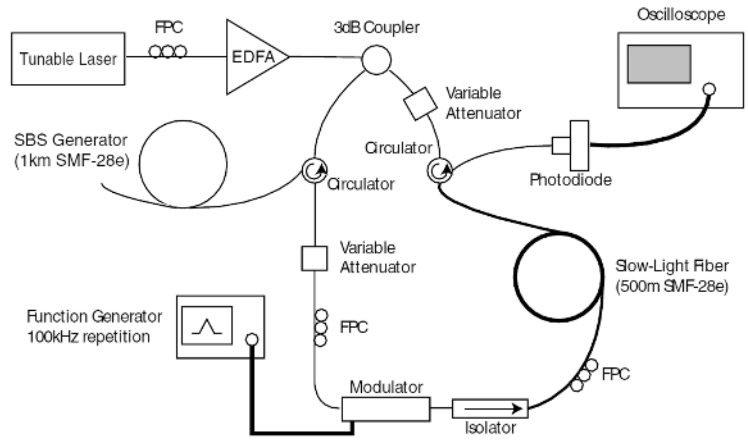

FIG. 2: Experimental setup for SBS slow light. (After Ref. [12].)

both the pump wave and probe pulses. The probe wave was created from a SBS generator pumped by the light from the laser. Probe pulses counter-propagate with respect to the pump wave in a 500-m-long SMF-28e fiber whereby the pulses experienced amplification and slow-light delay. The delayed output pulses were recorded by a fast detector and displayed on an oscilloscope. The experiment demonstrated that the delay can be tuned continuously by as much as 25 ns by adjusting the intensity of the pump field and that the technique can be applied to pulses as short as 15 ns. Figure 3 shows the measured slow-light delay as a function of the gain parameter for input pulsewidths of 63 ns and 15 ns. Figure 4 shows the delayed (solid line) and un-delayed (dotted line) pulses for both pulse widths at a gain parameter of $G = 11$. A fractional slow-light delay of 1.3 was achieved for the 15-ns-long input pulse with a pulse broadening of 1.4.

It is also possible to observe fast light (where $v_g > c$ or $v_g < 0$) due to SBS. In particular, anomalous dispersion occurs at the center of the absorbing resonance at the anti-Stokes frequency, which can be seen by repeating the analysis given above but with the sign of g inverted. Song *et al.* [14] used this effect to observe pulse advancement in an optical fiber.

Following the first demonstrations of SBS slow light in optical fibers, there has been considerable interest in exploiting the method for telecommunication applications. The appeal of the technique is that it is relatively simple, works at room temperature and at any wavelength where the material is transparent, and uses off the shelf telecommunication components.

One line of research has focused on reducing pulse distortion and improving system

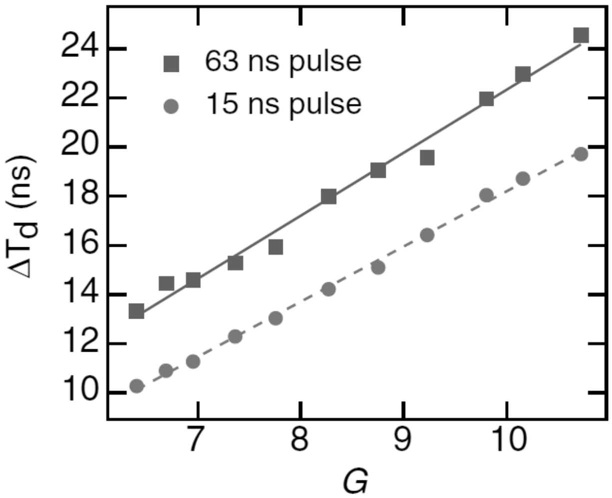

FIG. 3: Demonstration of optically controllable slow-light pulse delays. Induced delay as a function of the Brillouin gain parameter G for 63 ns long (square) and 15 ns long (circle) input Stokes pulses. (After Ref. [12].)

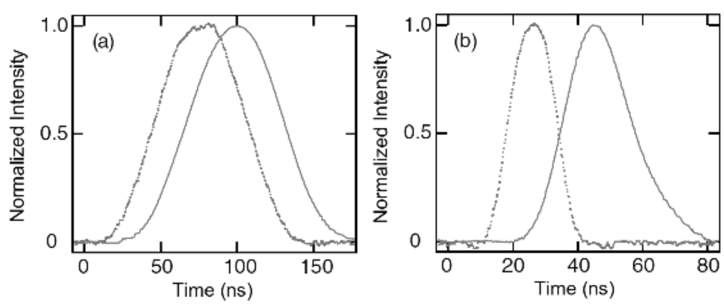

FIG. 4: Temporal evolution of the Stokes pulses (with a gain parameter $G = 11$) emitted from the fiber in the absence (dotted) and presence (solid) of the pump beam for (a) 63 ns long and (b) 15 ns long input Stokes pulses. (After Ref. [12].)

performance [15–21]. Stenner *et al.* were the first to consider this problem [15]. They proposed and demonstrated a general distortion management approach in which multiple SBS gain lines are used to optimize the slow light delay under certain distortion and pump power constraints. In their experimental demonstration with two closely-spaced SBS gain lines generated by a dual-frequency pump, approximately a factor of 2 increase in slow-light pulse delay was achieved as compared with the optimum single-SBS-line delay. The results suggested an effective way for reducing pulse distortion and improving SBS slow light by tailoring the gain resonance experienced by the signal pulses. Along this line, SBS slow-light efficiency (in terms of time delay per dB of gain) was also improved by customizing the gain resonance profiles using various approaches [22–24].

Another line of research has focused on broadband SBS slow light [25–31]. The width of the resonance which enables the slow-light effect limits the minimum duration of the optical pulse that can be effectively delayed without much distortion, and therefore limits the maximum data rate of the optical system. Fiber-based SBS slow light is limited to data rates less than a few tens of Mb/s due to the narrow Brillouin resonance width ($\sim$30 MHz in standard single-mode optical fibers). Herráez *et al.* were the first to increase the SBS slow-light bandwidth and achieved a bandwidth of about 325 MHz by broadening the spectrum of the SBS pump field. Zhu *et al.* extended this work to achieve a SBS slow-light bandwidth as large as 12.6 GHz, thereby supporting a data rate of over 10 Gb/s [26]. In this study, 75-ps data pulses were delayed by up to 47 ps at a gain of about 14 dB. Figure 5 shows the experimental results of the broadband SBS slow light. As can be seen in Fig. 5a, the SBS gain resonance occurring at $\omega_p - \Omega_B$ and the absorption resonance occurring at $\omega_p + \Omega_B$ are broadened to the point that they nearly overlap. In this situation, the anti-Stokes absorption tends to reduce the slow-light efficiency and limit the achievable bandwidth to about Ω_B. This limitation was recently overcome using a second broadband pump whose center frequency is higher than the first broadband pump by $2\Omega_B$ [28, 29]. This second pump generates a gain resonance that compensates the absorption caused by the first pump, enabling the use of even broader pump spectra to increase the SBS slow-light bandwidth. A bandwidth of about 25 GHz was achieved using this approach [29].

Other research in SBS slow light includes decoupling gain from delay [32–34], and reducing control latency [22, 35]. In SBS slow light, pulse delay is always companied by signal amplification. This amplification may not be desirable in situations where nearly constant signal powers are needed. To reduce the signal power variation in slow light, Zhu *et al.* used two widely-separated anti-Stokes lines to form a slow-light element that exhibits a less variation of signal amplitudes than in the single SBS resonance case [32]. In Refs. [33] and [34], a composite profile was created by combining a broadened absorption resonance and a narrow gain resonance. This composite profile was used to achieve slow-light delays with constant signal amplitude. Control latency in SBS slow light is determined by the smaller of the time it takes to adjust the pump laser intensity or twice the time it takes light to transit the fiber, with the latter often being a limiting factor. Therefore, use of a

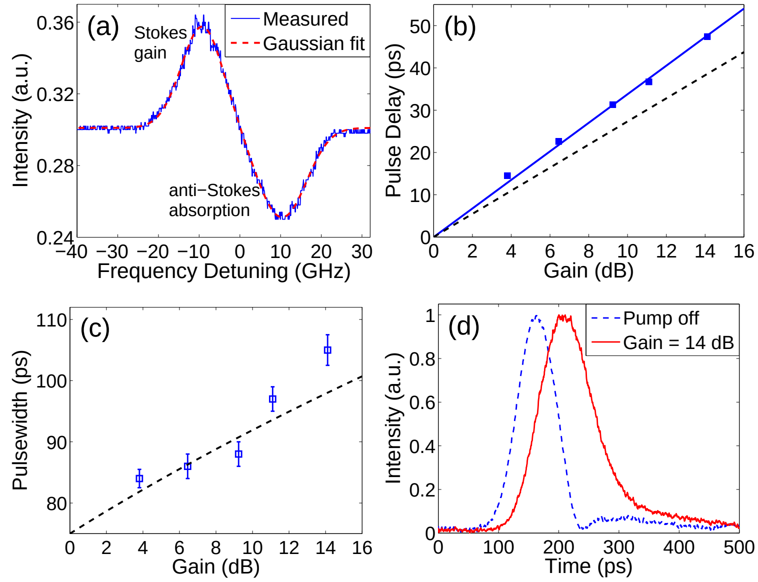

FIG. 5: (a) Measured SBS gain spectrum with a dual Gaussian fit. The SBS gain bandwidth (FWHM) is found to be 12.6 GHz. Pulse delay (b) and pulse width (c) as a function of SBS gain. In (b), the solid line is the linear fit of the measured data (solid squares), and the dashed line is obtained without considering the anti-Stokes absorption. In (c), the dashed curve is the theoretical prediction. (d) Pulse waveforms at 0-dB and 14-dB SBS gain. The input data pulsewidth is ∼75 ps. (After Ref. [27].)

short length of fiber for SBS slow light can increase the speed of adjusting SBS slow light delays [22, 35].

Additionally, non-silica fibers have attracted substantial interest in the slow-light research community. For example, chalcogenide glass fibers [22, 36] and tellurite glass fibers [37], which exhibit much larger SBS gain factors g_0, are being investigated for SBS slow light. A critical comparison of different fibers for SBS slow light is given in Ref. [36].

B. Slow light via SRS

Slow light via stimulated Raman scattering can also be achieved in optical fibers, but over much larger bandwidths and hence can be used with pulses of much shorter duration. The scattering arises from exciting vibrational or rotational motions in individual molecules – also known as optical phonons – as opposed to exciting sound waves as in the SBS process. Regardless of the microscopic mechanism for creating the material excitation, slow light via

SRS can be similarly understood by considering the resonances induced by a pump beam propagating in the fiber.

Compared to the simple Lorentzian shaped resonances in SBS, SRS in an optical fiber has a more complicated resonance shape, which depends on the fiber structure and material composition. Figure 6 shows the typical SRS spectral response function $\tilde{h}_R$ in a silica fiber [38, 39], where the largest gain peak of the Stokes band occurs about 13.2 THz below the pump beam frequency. The real part of $\tilde{h}_R$ is proportional to the refractive index change due to SRS, while the imaginary part is proportional to the gain (for the Stokes band) or loss (for the anti-Stokes band). Although the gain spectrum (*i.e.*, the imaginary part of $\tilde{h}_R$) is far from a Lorentzian shape, the real part of $\tilde{h}_R$ still has a nearly linear variation with frequency near the center of the band, suggesting that slow light is readily observable.

A model for the measured Raman response function consisting of a single damped oscillator has been proposed in Ref. [40] and is given by

$$h_R(t) = (\tau_1^2 + \tau_2^2)/(\tau_1 \tau_2^2) \exp(-t/\tau_2) \sin(t/\tau_1), \tag{7}$$

where $\tau_1 = 12.2$ fs and $\tau_2 = 32$ fs. Its Fourier transform $\tilde{h}_R$ is shown as thin lines in Fig. 6. As seen in the figure, the line shape is nearly Lorentzian and thus, to good approximation, the results presented above for SBS can be applied to SRS by replacing the gain coefficient and linewidth with those for SRS. The SRS linewidth is about 9.5 THz in this model.

As a resut of its broad linewidth, SRS is most useful for delaying picosecond-duration or shorter pulses. Because of the relatively small Raman gain coefficient in standard fibers, high power continuous-wave or pulsed pump beams are usually required to obtain delays that are a sizable fraction of the input pulse width. However, the propagation of picosecond pulses under strong pumping conditions necessitates an understanding of the roles of fiber dispersion, self-phase modulation, cross-phase modulation, and group velocity mismatch and their competition with the SRS process.

Another difference between SRS and SBS is that SRS can occur for both the copropagating and counterpropagating pump-probe geometries. In the case of counterpropagating pump and probe beams, a continuous-wave pump beam is usually required so that it overlaps substantially with the probe pulse as it propagates along the fiber. However, due to the lower threshold for SBS self-generation in an optical fiber (about 2 to 3 orders of magnitude lower than that of SRS), SBS self-generation will deplete the pump beam and suppress

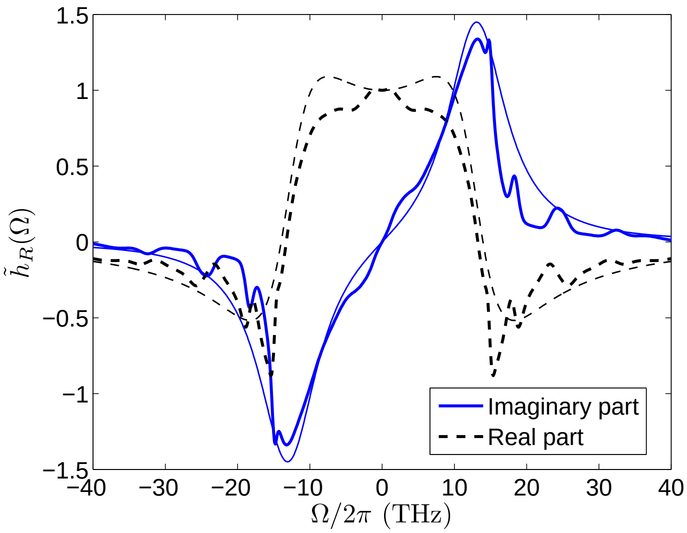

FIG. 6: The real and imaginary parts of the typical Raman gain spectrum $\tilde{h}_R(\omega)$ of a silica fiber [38, 39]. The thin lines are obtained by fitting the experimental observations to the simple damped oscillator model given in Ref. [40].

SRS. Therefore, most SRS experiments use copropagating pump and probe beams where the pump beam is pulsed with a duration shorter than the acoustic lifetime (typically a few nanoseconds). One additional consideration for generating large SRS gain and slow light delay is to match the group velocities of pump and probe pulses so that they do not separate temporally as they propagate through the fiber. It is possible to achieve a long walk-off length by selecting either a fiber with suitable dispersive properties or choosing a suitable pump wavelength for a given fiber.

Sharping *et al.* demonstrated an ultrafast all-optical controllable delay in a fiber Raman amplifier [41]. In this experiment a 430-fs pulse is delayed by 85% of its pulse width. The ability to accommodate the bandwidth of pulses shorter than 1 ps in a fiber-based system makes SRS slow-light useful for producing controllable delays in ultra-high bandwidth telecommunication systems. Figure 7 shows the experimental setup where copropagating signal and pump pulses are used. Signal pulses are derived from a Ti:Sapphire-laser-pumped optical parametric oscillator (OPO) and have a center wavelength of 1640 nm and a transform-limited temporal width of 430 fs. Fourier-transform spectral interferometry (FTSI) is used to measure the delay of the signal pulses since it provides the ability to accurately measure delays as small as tens of fs between pulses of very low peak power. Temporally-separated

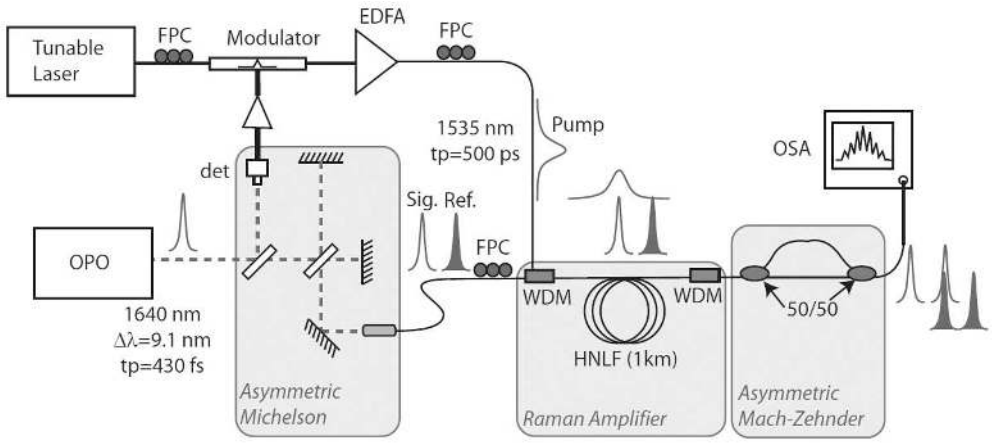

FIG. 7: The experimental setup used to demonstrate slow light in a Raman fiber amplifier. (After Ref. [41].)

reference pulses, which are used to measure the signal delay, are generated by passing the signal pulse train through an asymmetric Michelson interferometer. 500-ps-long, 1535-nm pump pulses are combined synchronously and copropagate with the signal and reference pulses in a 1-km-long highly nonlinear fiber (HNLF), where the signal pulse is amplified and delayed while the reference pulse is not amplified. The interferogram, as measured by an optical spectrum analyzer (OSA), is used to determine the slow light delay via Fourier transform.

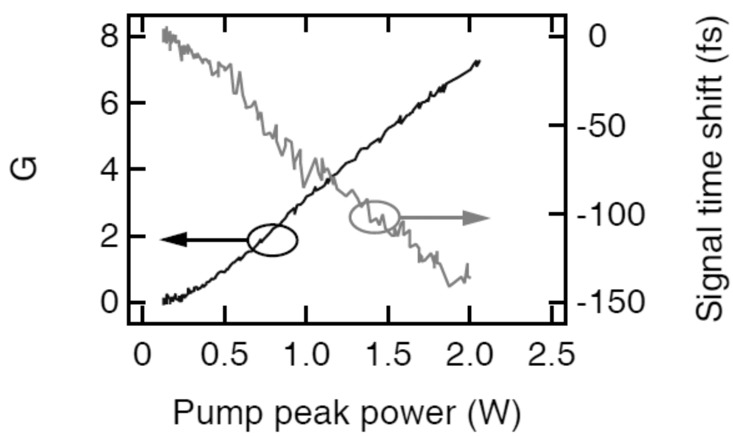

FIG. 8: Signal gain and delay vs. pump power in SRS slow light. (After Ref. [41].)

Figure 8 shows the slow-light results in the Raman amplifier where the signal wavelength is 1637 nm, which is very close to the peak of the Raman gain profile. The gain parameter (left axis) and signal time delay (right axis) are plotted as functions of pump peak power. It is seen that both the gain and delay vary linearly with pump power. The maximum delay achieved for the data depicted in Fig. 8 is 140 fs, which is 40% of the transform-limited input pulse width.

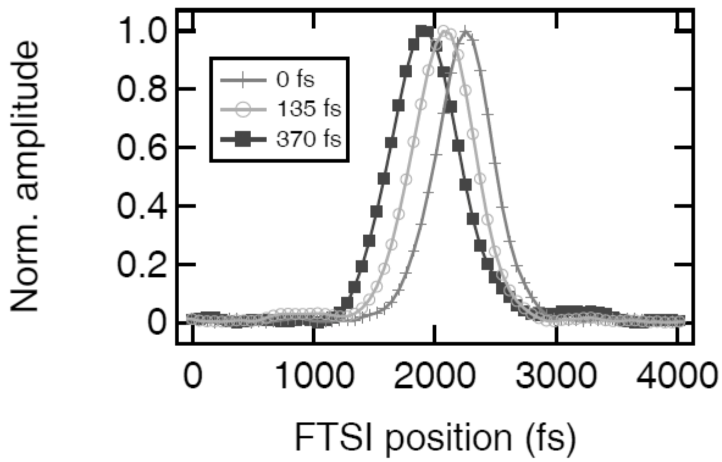

FIG. 9: Amplitude of the transformed interferograms for pulse delay changes of 0 fs, 135 fs and 370 fs. (After Ref. [41].)

Figure 9 shows three delayed signal pulses in the Fourier-transformed, time-domain representation, where the system is adjusted to obtain as large a delay as possible prior to each measurement. The measured delay of the pulse peak has contributions from SRS, cross-phase modulation, and wavelength shifts. For these pulses, the relative signal delay is varied from 0 to 370±30 fs, which is 85% of the input pulse duration at maximum delay. The measured gain for the case of maximum delay is $G = 7$, which implies a Raman gain bandwidth of 3 THz extrapolated from $\Delta T_d = G/\Gamma_R$. It is seen that the spectral width of the pulses does not change from the zero-delay case as the delay is varied.

In addition to optical fibers, SRS slow light has also been demonstrated in a silicon-on-insulator planar waveguide [42]. Here, a group-index change of 0.15 was produced using an 8-mm-long nanoscale waveguide, and controllable delays as large as 4 ps for input signal pulses as short as 3 ps was demonstrated. This scheme represents an important step in the development of chip-scale photonics devices for telecommunication and optical signal processing.

C. Coherent Population Oscillations

Coherent population oscillations (CPO) are a quantum effect that creates a narrow spectral hole in an absorption profile. The rapid variation of refractive index in the neighborhood of the spectral hole leads to slow or fast light (superluminal) propagation.

Coherent population oscillations occur when the ground-state population of a saturable medium oscillates at the beat frequency between a pump wave and a probe wave. The population oscillations are appreciable only for $\delta T_1 \lesssim 1$, where δ is beat frequency and T_1 is the ground state recovery time. When this condition is met, the pump wave can efficiently scatter off of the temporally modulated ground-state population into the probe wave, resulting in reduced absorption of the probe wave. In the frequency domain, this leads to a narrow spectral hole in the absorption profile, and the hole has a linewidth on the order of the inverse of the excited-state lifetime.

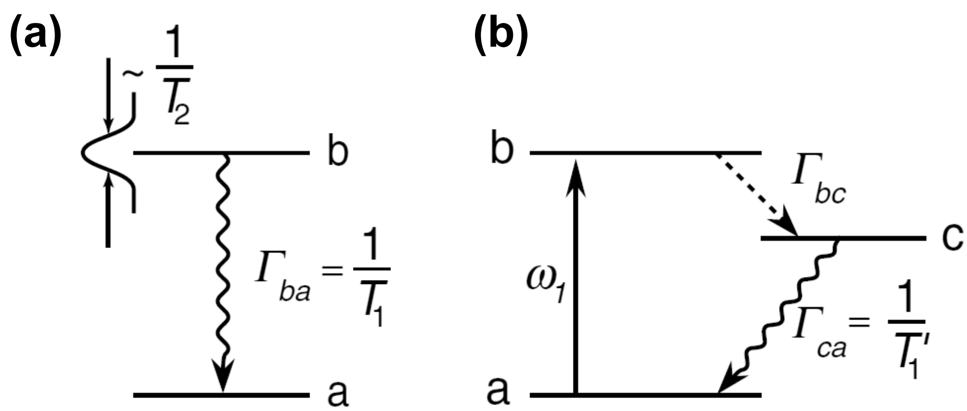

FIG. 10: (a) Two-level system for observing CPO. (b) Relevant energy levels in Ruby used for CPO slow light. (After Ref. [43].)

A thorough understanding of CPO can be obtained via the density matrix equations of motion for a simple two-level system (Fig. 10a). The shape of the probe absorption profile obtained using this formalism is given by

$$\alpha(\delta) = \frac{\alpha_0}{1+I_0}\left[1 - \frac{I_0(1+I_0)}{(T_1\delta)^2 + (1+I_0)^2}\right], \tag{8}$$

where α_0 is the unsaturated absorption coefficient, $I_0 = \Omega^2 T_1 T_2$ is the pump intensity normalized to the saturation intensity, Ω is the Rabi frequency, and T_2 is the dipole moment dephasing time. The refractive index variation associated with the absorption feature leads to a large increase of the group index and slow light pulse propagation. The slow-light delay at line center is given by

$$\Delta T_d = \frac{\alpha_0 L T_1}{2} \frac{I_0}{(1+I_0)^3}, \tag{9}$$

where L is length of the slow light medium. It is seen that there is no theoretical limit to the amount of delay achievable via CPO, since L is unbounded. In practice, however, residual absorption as well as group-velocity dispersion and spectral reshaping impose a limit to the achievable fractional delay to around 10%.

Spectral holes due to CPO were first predicted in 1967 by Schwartz and Tan [44]. The first experimental investigation of spectral holes caused by CPO observed an extremely narrow transmission window (37 Hz width) in a ruby absorption band near a wavelength of 514.5 nm [45]. Slow light using this narrow spectral feature was first demonstrated by Bigelow *et al.* [43], where υ_g as low as 57.5 ± 0.5 m/s was observed in a 7.25-cm-long ruby crystal at room temperature (the relevant energy levels are shown in Fig. 10b). Slow light propagation was also demonstrated in an alexandrite crystal at room temperature using CPO [46], in semiconductor structures [47–50], and erbium-doped fibers [51, 52].

In CPO slow light experiments, the pump and probe need not be separate beams; a single beam with a temporal modulation can experience slow-light delay. However, the modulation frequency or the pulse spectral width should be narrow enough to essentially fit within the spectral hole for the slow light effect to be appreciable with minimum pulse distortion. This means that the slow light bandwidth is limited by the width of the spectral hole created by CPO.

In addition to an absorbing medium, CPO can also occur in an amplifying (population inverted) medium. In this case, a spectral hole is created in a gain feature, and the resulting anomalous dispersion can lead to superluminal or negative group velocities. Both slow- and fast-light effects were recently demonstrated in Erbium-doped fibers where the absorption or gain can be controlled by a pump laser that creates a population inversion [51, 52]. In one experiment [51], modulated or pulsed light at a wavelength of 1550 nm was delayed or advanced in 13-m-long Erbium-doped fiber that was pumped by a counter-propagating pump laser operating at a wavelength of 980 nm. A maximum fractional advancement of 0.12 and a maximum fractional delay of 0.09 were achieved. Figure 11 shows the frequency and pump power dependence of the fractional delay observed in propagation through the erbium-doped fiber, where the regime of operation can be tuned by adjusting the pump power. Figure 12 shows the fractional advancement as a function of pulsewidth in both the slow light and superluminal light regimes.

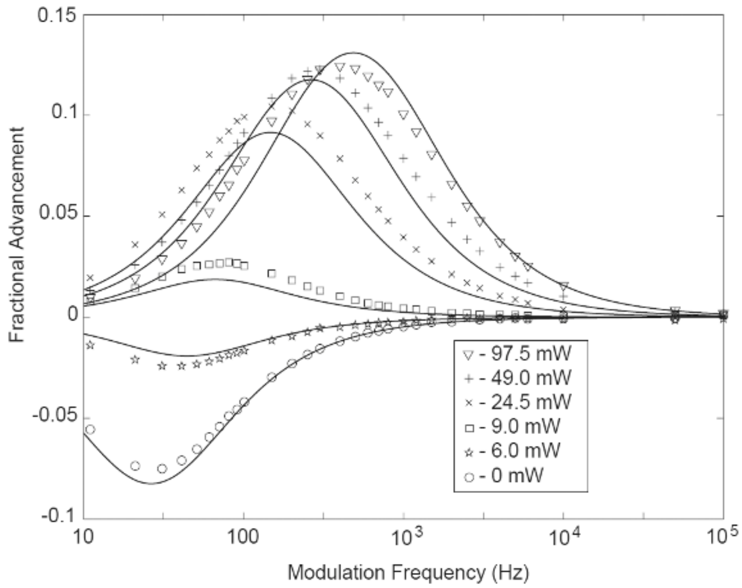

FIG. 11: Frequency and pump power dependence of the fractional delay observed in propagation through erbium-doped fiber. The input was sinusoidally modulated beam at 1550 nm with an average power of 0.8 mW. Results of the numerical model are shown as solid lines along with the experimental data points. (After Ref. [51].)

The superluminal light propagation with negative group velocity due to CPO has been recently observed in an Erbium-doped fiber where the signal pulse appears to propagate backwards [52], which demonstrates that "backwards" propagation is a realizable physical effect.

Like other slow- and fast-light techniques discussed so far, CPO based slow- and fast-light pulse propagation also suffers from pulse distortion. In the fast light regime, pulse broadening and compression in an EDFA can be described in terms of two competing mechanisms: gain recovery and spectral broadening [53]. Shin *et al.* observed that pulse distortion caused by these effects depends on input pulse width, pump power, and background-to-pulse power ratio. They demonstrated that pulse distortion can be minimized by a proper choice of the these parameters [53]. In their experiment, significant fractional advancement ($\sim$0.17) and minimal distortion was obtained for a background-to-pulse power ratio of $\sim$0.75, a pump power of 17.5 mW, and an input pulse duration of 10 ms.

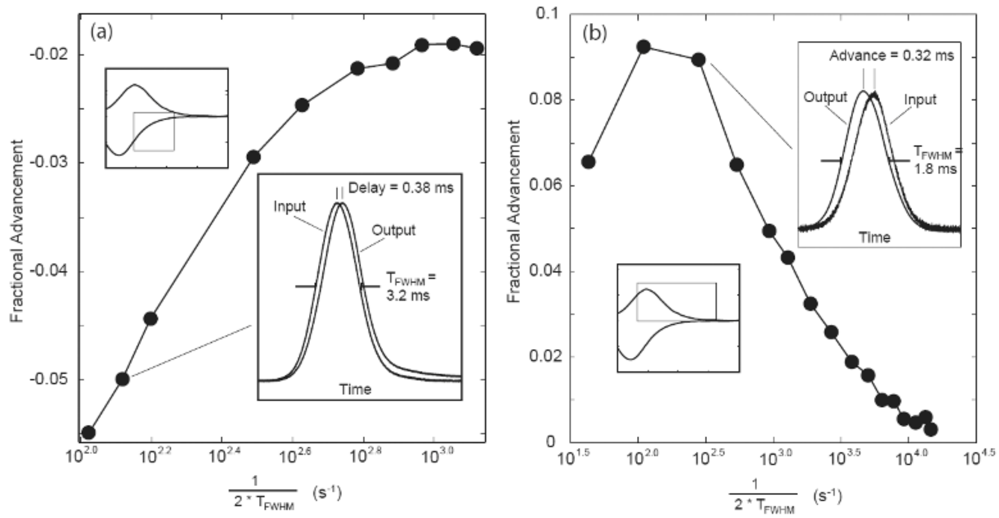

FIG. 12: Fractional advancement vs. the log of the inverse pulse width in the regimes of slow light (a) and superluminal propagation (b). (After Ref. [51].)

Although the bandwidth of CPO slow or fast light in erbium-doped fibers is typically limited to the kHz frequency range, CPO slow light in semiconductors exhibits bandwidths exceeding 1 GHz [47, 48]. This large bandwith, together with mature semiconductor processing techniques, makes CPO in semiconductor structures an important route to achieve chip-scale slow-light devices.

D. Electromagnetically Induced Transparency in Hollow-Core Fibers

As mentioned in the introduction to this Chapter, electromagnetically induced transparency (EIT) [54] is a technique that renders an atomic medium transparent over a narrow spectral range within an absorption band. It has been observed in atomic vapors [55], Bose-Einstein condensates [2], crystals [4], and semiconductor quantum wells and quantum dots [47]. Applications of EIT include ultraslow light [2, 3], stored light [56, 57], enhancement of nonlinear optical effects [58], and quantum information processing [59].

In its simplest form, EIT can take place in a three-level Λ system as shown schematically in Fig. 13. A weak probe field (ω_p) is tuned near the $|1\rangle \leftrightarrow |2\rangle$ transition frequency and is used to measure the absorption spectrum of the transition, while a much stronger coupling field (ω_c) is tuned near the $|2\rangle \leftrightarrow |3\rangle$ transition frequency. The $|1\rangle \leftrightarrow |3\rangle$ transition is dipole forbidden. Quantum interference between the $|1\rangle \leftrightarrow |2\rangle$ and $|3\rangle \leftrightarrow |2\rangle$ transition

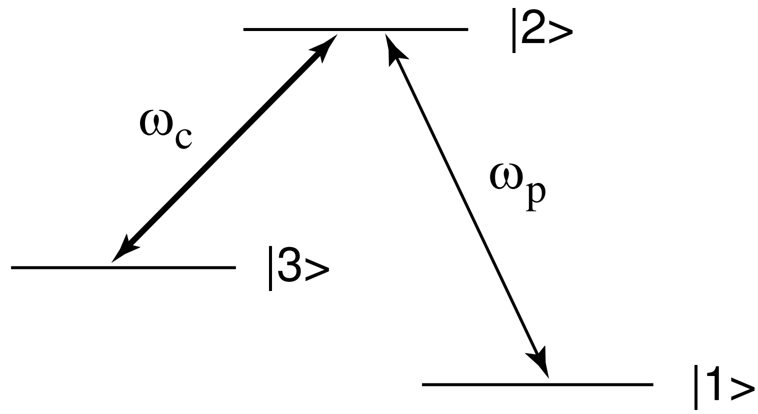

FIG. 13: EIT in a three-level Λ system.

amplitudes result in a cancellation of the probability amplitude for exciting state $|2\rangle$, thereby reducing the probe beam absorption. If the state $|3\rangle$ has a long lifetime, the quantum interference results in a narrow transparency window completely contained within the $|1\rangle \leftrightarrow |2\rangle$ absorption line. The rapid change in refractive index in the narrow transparency window produces an extremely low group velocity for the probe field, which leads to slow light.

Under appropriate conditions [60, 61], the transparency window created by EIT is approximately Lorentzian, whereby the frequency-dependent complex absorption coefficient is given by

$$\alpha(\delta) = \alpha_0(1 - \frac{f}{1 + \delta^2/\gamma^2}), \tag{10}$$

with

$$f = \frac{|\Omega_c/2|^2}{\gamma_{31}\gamma_{21} + |\Omega_c/2|^2}, \gamma = \frac{|\Omega_c/2|^2}{\gamma_{21}}, \tag{11}$$

where Ω_c is the Rabi frequency of the strong coupling field, γ_{21} is the coherence dephasing rate of the $|1\rangle \leftrightarrow |2\rangle$ transition, and γ_{31} is the dephasing rate of the ground state coherence. The steep, linear region of the refractive index in the center of the transparency window gives rise to slow light. The slow light delay at the center of the window is given by

$$\Delta T_d = \frac{\alpha_0 L}{2} \frac{\gamma_{21}}{\gamma_{31}\gamma_{21} + |\Omega_c/2|^2}. \tag{12}$$

There is no theoretical upper limit to the amount of delay that can be created by EIT as in the CPO case, while EIT has a practical advantage over CPO in that the transparency depth f can often approach 1, greatly reducing residual absorption. However, spectral reshaping limits the fractional advancement to approximately

$$\Delta T_{frac} = \frac{3}{2} \frac{|\Omega_c/2|^2 \, T_{\mathrm{FWHM}}}{\gamma_{21}}. \tag{13}$$

Slow light based on EIT has been demonstrated in various material systems. For example, Hau *et al.* observed a group velocity of 17 m/s in a Bose-Einstein condensate [2] and Turukhin *et al.* demonstrated the propagation of slow light with a velocity of 45 m/s through a solid crystal at a cryogenic temperature of 5 K [4].

While these results are truly impressive, there has been increased interest in achieving EIT in waveguides where coherent light-matter interactions are enhanced due to tight light confinement and long interaction lengths. For example, integrated hollow-core ARROW waveguides filled with an atomic vapor were recently proposed and investigated as a platform for chip-scale EIT [62–64]. It is also possible to observe EIT effects by placing an atomic medium outside of a waveguide. For example, Patnaik *et al.* proposed a fiber-based EIT slow-light scheme where a tapered fiber is immersed in an atomic vapor, where they predict theoretically that very slow group velocities are possible [65].

Another approach is to fill a hollow-core photonic-bandgap fiber (HC-PBF) with a gas. These fibers are of great interest because the light can be confined and guided with low loss in a hollow core surrounded by a photonic crystal structure that localizes light in the core. By filling the hollow core with desirable gases, resonant optical interactions, such as EIT, can be achieved over a long interaction length. Such a HC-PBG gas cell is compact and can be integrating with existing fiber-based technologies.

EIT-based slow-light propagation has recently been demonstrated experimentally in a HC-PBF filled with acetylene [66]. Figure 14 shows the experimental setup for EIT slow light in a HC-PBF fiber and a scanning electron microscope image of the transverse structure of the fiber. The 1.33-m-long HC-PBF has a core diameter of 12 μm and a band gap extending from 1490 nm to 1620 nm. The fiber ends were sealed in vacuum cells, and the fiber core was filled with 99.8% pure acetylene (C_2H_2). The energy-level scheme for the molecules used in the experiment is shown in Fig. 15. The probe beam and the coupling beam are tuned to the R(15) and P(17) lines of $^{12}C_2H_2$ at 1517.3144 nm and 1535.3927 nm, respectively.

Figure 15(a) shows the 480-MHz-wide Doppler-broadened absorption spectrum of in the absence of the control beam. In the presence of the control beam, a transparency window is opened. Figure 15(b) show a typical trace of the probe-field absorption in presence of a

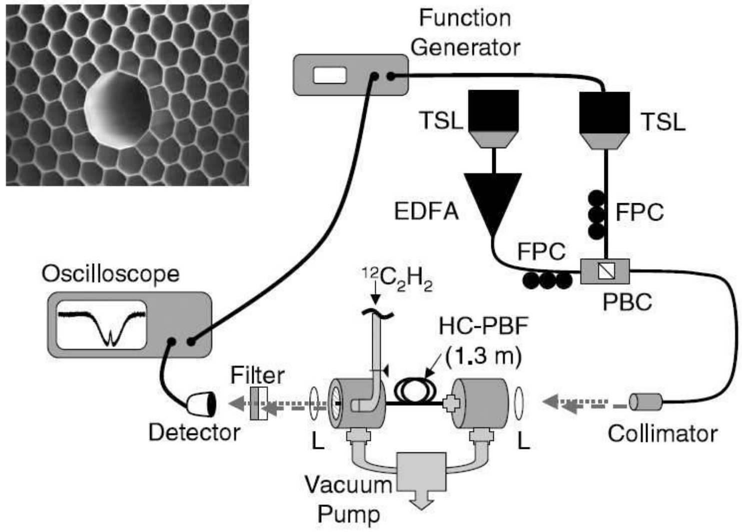

FIG. 14: Experimental setup for EIT slow light in a hollow-core fiber. (After Ref. [66].)

320-mW (measured at the output of the fiber) control beam tuned exactly to the center of the P(17) transition. The observed transparency feature in acetylene is remarkable in that the transition strength is much weaker than transitions of gases commonly used in EIT.

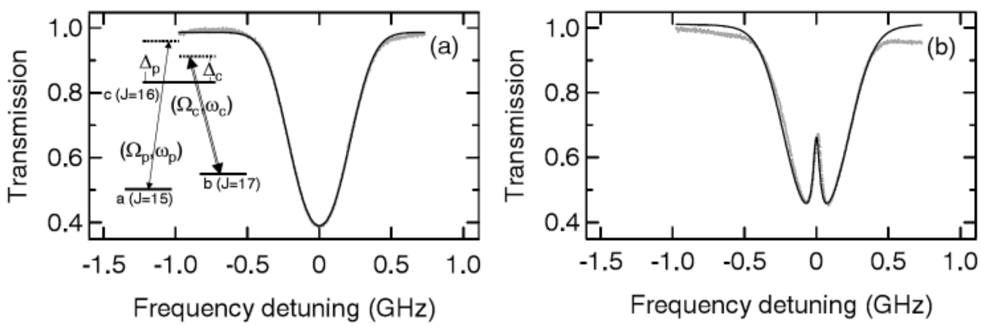

FIG. 15: Measured absorption for the probe beam. (After Ref. [66].)

The transparency window can be tuned by the control beam. Figure 16(a) shows the measured transparency as a function of the control power measured at the output end of the fiber together with the corresponding theoretical prediction. The measured transparency full-width at half maximum (FWHM) in Fig. 16(b) shows a linear dependence on control power.

Slow light was demonstrated by propagating a 19-ns-long probe pulse through the acetylene-filled hollow-core fiber. Figure 17 shows the output probe pulse with control

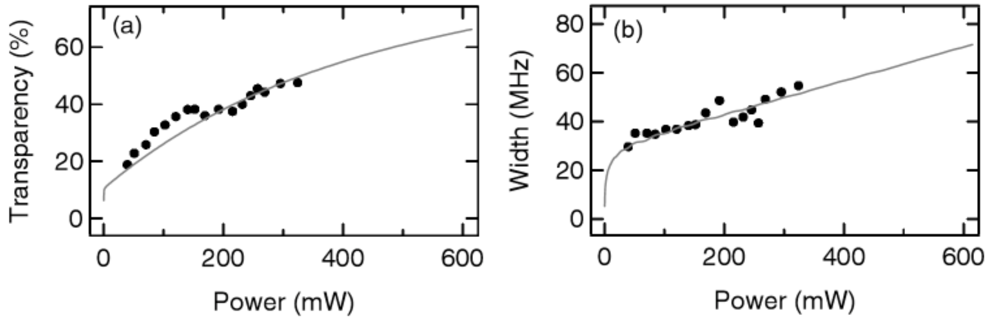

FIG. 16: Transparency and bandwidth of EIT. (After Ref. [66].)

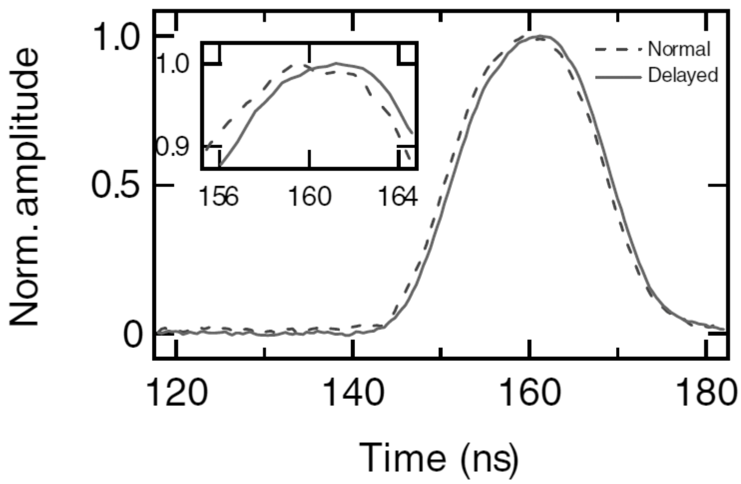

FIG. 17: Observation of slow light via EIT in acetylene-filled hollow-core fiber. (After Ref. [66].)

beam on and off. The delay measured in the presence of the control beam is 800 ps. This is the first demonstration of EIT slow light at telecommunication wavelengths.

Besides acetylene, alkali atomic vapors such as Rubidium have also been injected into a HC-PBF to achieve ultra-low power level optical interactions [67]. Alkali atomic vapors are a favorite choice for EIT because of the relatively simple energy-level structure and large atomic absorption cross section that can result in a large optical depth for modest atomic number densities. However, due to the strong interaction of the alkali atoms with the walls of the silica fiber, making a useful HC-PBF-based alkali vapor cell has been a challenge. By coating the inner walls of the fiber core with organosilane and using light-induced atomic desorption to release alkali atoms into the core, Ghosh *et al.* realized a significant rubidium density in the core of a HC-PBF with an optical depth over 2,000 [67]. They demonstrated EIT in such a HC-PBF Rubidium cell with a control power as low as

10 nW, which represents more than a factor of 10^7 reduction in the required control power in comparison to that needed to achieve EIT in the HC-PBF acetylene cells. These results suggest that EIT in hollow-core fibers is a viable and important route to slow light.

E. Wavelength Conversion and Dispersion

The slow light methods discussed above rely on the use of either absorbing or amplifying resonances, which give rise to rapid variations in refractive index within a narrow spectral range. Dramatic slow-light delays can also be obtained by a method known as wavelength-conversion-and-dispersion that does not rely on such resonances. Instead, this method makes use of group velocity dispersion of an optical waveguide and wavelength conversion techniques. In essence, the incident signal pulses are converted to another carrier frequency via a wavelength conversion device, propagate through a length of highly-dispersive waveguide (with large group velocity dispersion), and the delayed pulses are converted back to the original wavelength via a second wavelength conversion device. Tunable group delay of signal pulses are then obtained because of the frequency dependence of the group velocity in the dispersive waveguide. The group delay as a function of the carrier wavelength shift $\Delta\lambda$ is given by

$$\Delta T_d = LD\Delta\lambda, \tag{14}$$

where L is the length of the highly-dispersive waveguide, D is the group velocity dispersion (GVD) parameter of the waveguide at the incident pulse carrier frequency, and $\Delta\lambda$ is the wavelength shift of the first wavelength conversion device.

This technique was first demonstrated in an optical fiber by Sharping *et al.* [68]. Figure 18 shows the schematic of the experiment setup, where wavelength conversion and re-conversion are accomplished by using a fiber-based parametric amplifier. The wavelength conversion and re-conversion happen in the same fiber (HNL-DCF) but in opposite propagation directions. The dispersive fiber (DCF) has a GVD of -74 ps/nm at the signal wavelength of 1565 nm. The signal pulses from an OPO have a pulsewidth of 10 ps and a repetition rate of 75 MHz. Tunable pulse delays in excess of 800 ps were demonstrated by varying the pump wavelength of the fiber-based parametric amplifier, yielding a relative delay of 80 pulsewidths. Figure 19 shows the measured delay and the corresponding temporal pulse shapes. Figure 20 shows the optical spectra of the input and the delayed pulses. The delayed

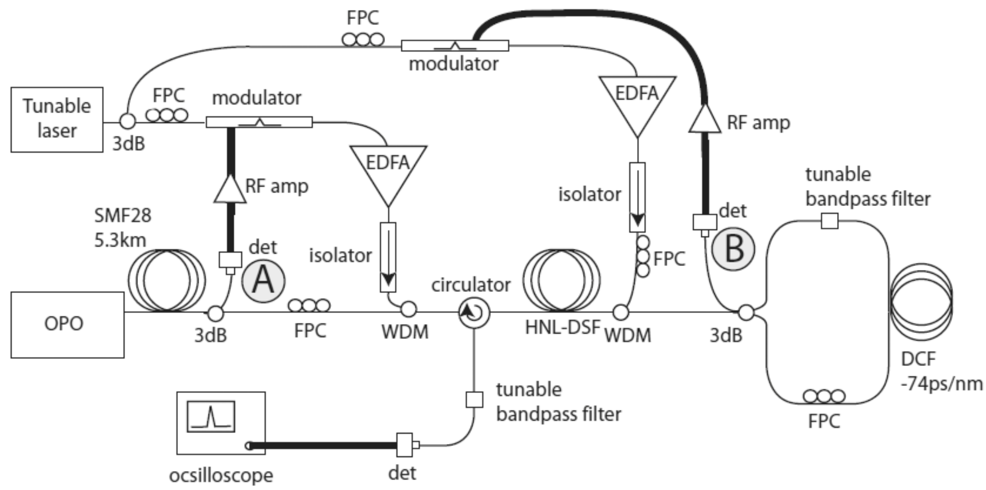

FIG. 18: Experimental setup for tunable delays via wavelength conversion and dispersion. (After Ref. [68].)

pulses have the same center wavelength as the input pulses as a result of the use of same frequency pump in the conversion and re-conversion in the same fiber. The spectral shape is nearly identical down to 8 dB from the peak, at which point considerable noise are observed in the sidebands. This noise is comprised of a combination of amplified spontaneous emission from the erbium-doped fiber and parametric amplifiers.

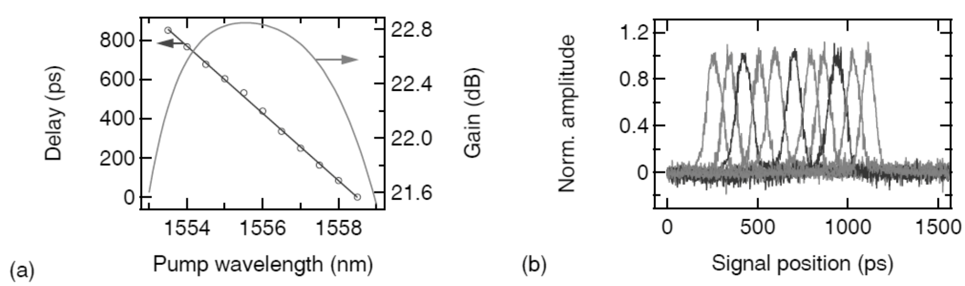

FIG. 19: (a) Measured delay and calculated gain versus pump wavelength. (b) Measured pulse traces at different pump wavelengths indicted in (a). (After Ref. [68].)

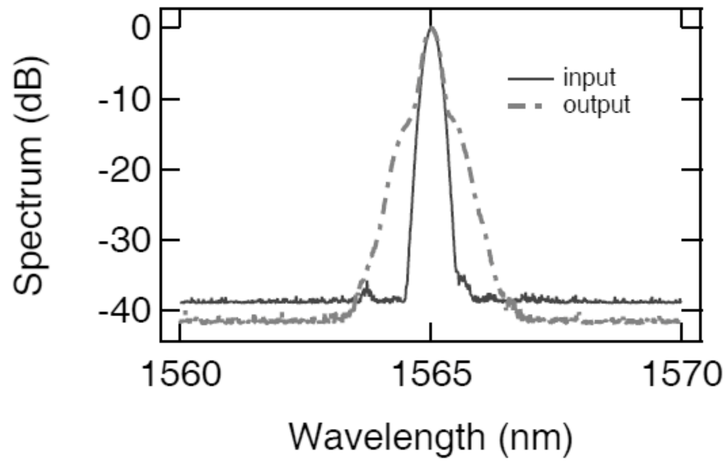

FIG. 20: Spectra of input and output pulses in one implementation of the conversion and dispersion process. (After Ref. [68].)

Okawachi *et al.* recently simplified the fiber-based wavelength-conversion-and-dispersion technique and delayed 3.5-ps pulses up to 4.2 ns, corresponding a fractional delay of 1200 [69]. Figure 21 shows the experiment setup where the wavelength conversion and re-conversion are achieved with self-phase-modulation spectral broadening followed by spectral filtering. Figure 22 shows the measured delay as a function of the center wavelength of the tunable filter. By suitable choice of the tunable filter, the spectrum of the delayed pulse can be made identical to that of the incident pulse.

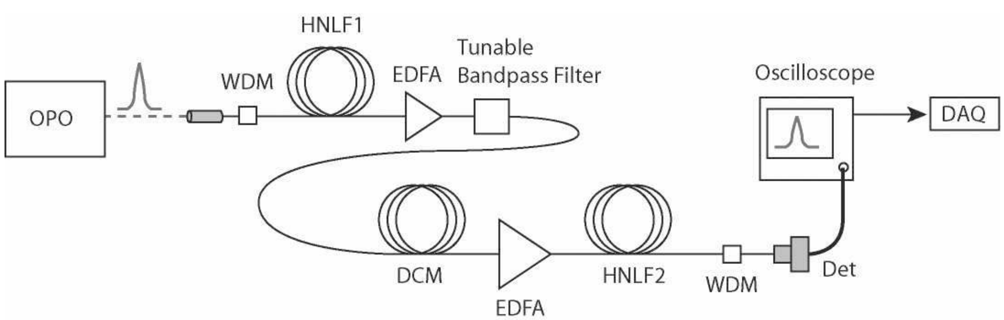

FIG. 21: Experimental setup. (After Ref. [69].)

Another improved variation of the conversion-dispersion technique uses a periodically poled lithium-niobate (PPLN) waveguide for wavelength conversion [70]. The dispersion element is a dispersion-compensated fiber (DCF), and intrachannel dispersion arising from the DCF is compensated by a chirped fiber Bragg grating after the second wavelength

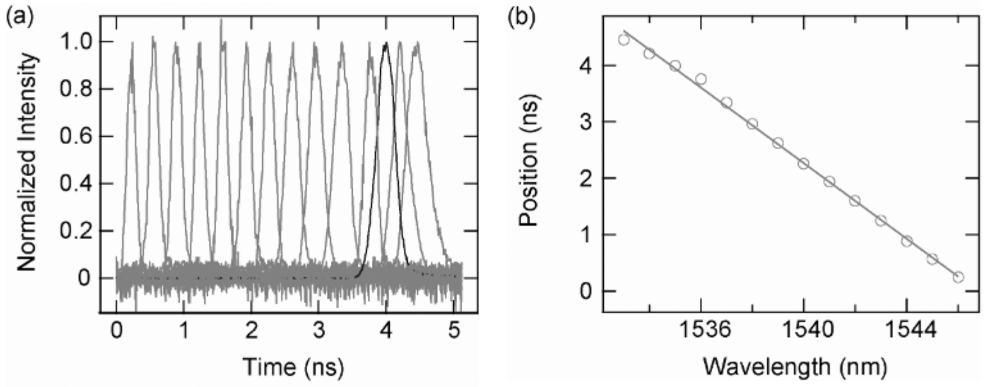

FIG. 22: (a) Measured pulse traces at different pump wavelengths. (b) Measured pulse delay versus pump wavelength. (After Ref. [69].)

conversion unit. The PPLN wavelength converter has the advantage of being rapidly tunable across a large bandwidth in a two-pump configuration, allowing both large delay variations and fast reconfiguration rates. In this experiment, continuous optical delay up to 44-ns is demonstrated for 10-Gb/s NRZ system, which is equal to 440-bit slots. As recently demonstrated in Ref. [71], the conversion-dispersion technique can be further improved using a silicon waveguide for the wavelength conversion stage, yielding an all-silicon integrated-circuit approach to slow light.

Compared with resonance-based slow-light techniques, the conversion-dispersion method has several advantages: 1) a highly controllable span of tunable delays from ps to ns and large fractional delay, 2) can support broad bandwidths suitable for data rates exceeding 10 Gb/s, and 3) the delayed pulses can have identical wavelength and bandwidth. Still, the reconfiguration rate is limited by the tuning speed of the filters or of the pump laser frequencies. Also, care must be used in choosing the dispersive element and wavelength conversion range to minimize dispersive pulse broadening and maximize slow light delay.

II. CONCLUSIONS

Slow light is an important research area that is of fundamental interest and has great potential for applications. The use of optical waveguides is particularly attractive for application since it can be readily integrated with existing technologies. In this Chapter, we have described several important developments in achieving all-optical tunable slow-light delay in optical fibers and other types of waveguides; we anticipate that some of these schemes will soon find their way into telecommunication systems, ultra-high-speed diagnostic equipment, and synthetic aperture RADAR and LADAR systems.

Acknowledgments

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